REPORT
GEOTECHNICAL STUDY
PROPOSED MILLVILLE KENNEL
4200 SOUTH 600 EAST
MILLVILLE, UTAH

November 13, 2024

Job No. 1342-001-24

Prepared for:
Cushing Terrell
800 West Main Street, Suite 800
Boise, Idaho 83702

Prepared by:
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November 13, 2024
Job No. 1342-001-24

Cushing Terrell
800 West Main Street, Suite 800
Boise, Idaho 83702

Attention: Mr. Scott Roberts

Ladies and Gentlemen:

Re: Report
Geotechnical Study
Proposed Millville Kennel
4200 South 600 East
Millville, Utah

1. INTRODUCTION

1.1 GENERAL

This report presents the results of our geotechnical study performed at the site of the proposed Millville Kennel, which will be located at 4200 South 600 East in Millville, Utah. The general location of the site with respect to major topographic features and existing facilities, as of 1998, is presented on Figure 1, Vicinity Map. A detailed location of the site showing existing roadways and surrounding facilities, on an air photograph base, is presented on Figure 2, Area Map. The locations and alignments of photographs taken of the site during the field portion of study and the test pit locations excavated in conjunction with this study are also presented on Figure 2.

1.2 OBJECTIVES AND SCOPE

The objectives and scope of our study were planned in discussions between Mr. Scott Roberts of Cushing Terrell and Mr. Patrick Emery of Gordon Geotechnical Engineering, Inc. (G^2).

In general, the objectives of this study were to:

1. Accurately define and evaluate the subsurface soil and groundwater conditions across the site.

2. Provide appropriate foundation, earthwork, geoseismic, and pavement recommendations to be utilized in the design and construction of the proposed structure.

In accomplishing these objectives, our scope has included the following:

1. A field program consisting of the excavating, logging, and sampling of 2 test pits which extended to depths of 7.5 to 10.5 feet below the existing grade.

2. A laboratory testing program.

3. An office program consisting of the correlation of available data, engineering analyses, and the preparation of this summary report.

1.3 AUTHORIZATION

Authorization was provided by returning a signed copy of our Professional Services Agreement No. 24-0814 dated August 14, 2024 and executed on September 30, 2024.

1.4 PROFESSIONAL STATEMENTS

Supporting data upon which our recommendations are based are presented in subsequent sections of this report. Recommendations presented herein are governed by the physical properties of the soils encountered in the exploration test pits for this study, measured and projected groundwater conditions, and the layout and design data discussed in Section 2., Proposed Construction, of this report. If subsurface conditions other than those described in this report are encountered and/or if design and layout changes are implemented, G^2 must be informed so that our recommendations can be reviewed and amended, if necessary.

Our professional services have been performed, our findings developed, and our recommendations prepared in accordance with generally accepted engineering principles and practices in this area at this time.

2. PROPOSED CONSTRUCTION

A kennel structure covering a footprint of 6,000 square feet is planned for the site. The proposed structure is anticipated to be one level above grade, and of light gauge steel-frame construction established slab-on-grade.

It is anticipated that the maximum column loads will be in the range of 60 to 90 kips. There may also be some internal bearing walls with loads in the range of 2 to 3 kips per lineal foot.

We estimate that maximum cuts and fills to achieve design grades will be minimal, on the order of two to three feet.

Paved entryways and at-grade parking surrounding the proposed building will also be a part of the overall development. Traffic over the entryway pavement surface will consist of a moderate volume of automobiles and light trucks, and a light volume of medium- and heavy-weight trucks. Traffic in the parking areas will be lighter.

3. INVESTIGATIONS

3.1 FIELD PROGRAM

In order to define and evaluate the subsurface soil and groundwater conditions across the site, 2 test pits were excavated to depths of 7.5 to 10.5 feet below existing grade. Locations of the test pits are presented on Figure 2.

The field portion of our study was under the direct control and continual supervision of an experienced member of our geotechnical staff. During the course of the excavation operations, a continuous log of the subsurface conditions encountered was maintained. In addition, relatively undisturbed and small disturbed samples of the typical soils encountered were obtained for subsequent laboratory testing and examination. The soils were classified in the field based upon visual and textural examination. These classifications have been supplemented by subsequent inspection and testing in our laboratory. Detailed graphical representation of the subsurface conditions encountered is presented on Figures 3A and 3B, Log of Test Pits. Soils were classified in accordance with the nomenclature described on Figure 4, Unified Soil Classification System.

Disturbed bag samples were collected from the soils brought up by the backhoe bucket.

Following completion of excavating and logging, each test pit was backfilled. Although an effort was made to compact the backfill with the backhoe, backfill was not placed in uniform lifts and compacted to a specific density. Consequently, settlement of the backfill with time is likely to occur.

Following completion of excavating operations, one and one-quarter-inch diameter slotted PVC pipe was installed in Test Pits TP-1 and TP-2 in order to provide a means of monitoring the groundwater fluctuations.

GORDON GEOTECHNICAL ENGINEERING, INC.

3.2 LABORATORY TESTING

3.2.1 General

In order to provide data necessary for our engineering analyses, a laboratory testing program was performed. The program includes moisture and partial gradation tests. The following paragraphs describe the tests and summarize the test data.

3.2.2 Moisture Tests

To aid in classifying the soils and to help correlate other test data, moisture tests were performed on selected undisturbed samples. The results of these tests are presented on the test pit logs, Figures 3A and 3B.

3.2.3 Partial Gradation Tests

To aid in classifying the soils and to provide general index parameters, a partial gradation test was performed upon three representative samples of the soils encountered in the exploration test pits. The test results are tabulated below:

Test Pit No.	Depth (feet)	Percent Passing No. 4 Sieve	Percent Passing No. 200 Sieve	Soil Classification
TP-1	4.0	17.2	6.2	GP/GM
TP-1	7.0	28.4	2.2	SP
TP-2	4.0	49.0	4.7	GP/SP

4. SITE CONDITIONS

4.1 SURFACE

At the time of the site visit, the site was mostly vacant and undeveloped. The overall site is part of the Millville Predator Research Facility. Vegetation consists of a light growth of grasses and weeds. Earthwork operations have previously occurred at the site.

The site grade generally slopes gently downward to the west with an overall relief of approximately three to five feet.

The site is bordered by existing structures to the north and south, similar undeveloped land to the east, and 600 East Street to the west.

Representative photographs of the site area are shown on Figure 5, Photographs.

4.2 SUBSURFACE SOIL

In order to define and evaluate the subsurface soil and groundwater conditions at the site, 2 test pits were excavated to depths of 7.5 to 10.5 feet below existing grade. Subsurface conditions encountered at the test pit locations were relatively consistent. At each of the test pit locations, loose/disturbed surficial soil was encountered to depths ranging from four to five inches. The surficial soil appeared to have been disturbed by past site grading/earthwork activity.

In general, underlying the loose/disturbed surficial soil in each of the test pits and extending to depths explored of 7.5 to 10.5 feet, natural soils were encountered. The natural soils consist primarily of gravel with varying amounts of sand and silt. The natural granular soils are medium dense, slightly moist, and brown and will exhibit high strength and low compressibility characteristics under the anticipated loading range.

The lines designating the interface between soil types on the test pit logs generally represent approximate boundaries. In-situ, the transition between soil types may be gradual.

4.3 GROUNDWATER

Groundwater was not encountered in the test pits to the depths explored, 7.5 to 10.5 feet, during excavating operations. Groundwater is anticipated to be at a depth greater than 20 feet at the site.

Seasonal and longer-term groundwater fluctuations of one to two feet should be anticipated. The highest seasonal levels will generally occur during the late spring and summer months.

5. DISCUSSIONS AND RECOMMENDATIONS

5.1 SUMMARY OF FINDINGS

Based on the results of our investigation, the proposed structure may be supported upon conventional shallow spread and wall foundations established on suitable natural soils.

The most significant geotechnical aspect of the site is the loose, surficial soils encountered to depths of four to five inches at each of the test pit locations. Loose, surficial soils must be completely removed from below the building footprint and exterior flatwork areas. The natural granular soils encountered at four to five inches are suitable for footing support.

Although not encountered at the test pit locations, non-engineered fills associated with the previous earthwork are likely to be encountered.

Due to the loose, surficial soils encountered and potential for encountering non-engineered fills, a qualified geotechnical engineer from our staff must aid in verifying that suitable natural granular soils have been encountered prior to the placement of structural site grading fills, floor slabs, footings, or foundations.

Detailed discussions pertaining to earthwork, foundations, floor slabs, lateral resistance, and the geoseismic setting of the site are discussed in the following sections.

5.2 EARTHWORK

5.2.1 Site Preparation

Initial preparation of the site must consist of the removal of the existing structures and pavements, debris, and any associated non-engineered fills. In proposed flexible pavement areas, the existing asphalt concrete and fills may remain provided that they do not interfere with the final grade. The asphalt concrete should be perforated to facilitate drainage and proofrolled.

Further preparation of the site must consist of the removal of all non-engineered fills, loose surficial soils, topsoil, debris, and other deleterious materials from beneath an area extending at least three feet beyond the perimeter of the proposed building, rigid pavement, and exterior flatwork areas.

The loose/disturbed surficial soil/non-engineered fills may remain in flexible pavement areas as long as they are properly prepared. Proper preparation will consist of scarifying and moisture conditioning the upper 9 to 12 inches and recompacting to the requirements of structural fill. However, it should be noted that compaction of fine-grained soils (clays and silts) as structural site grading fill will be very difficult, if not impossible, during wet and cold periods of the year. As an option for proper preparation and recompaction, the upper eight inches may be removed and replaced with granular subbase over proofrolled subgrade. Even with proper preparation, flexible pavements established on non-engineered fills may experience some long-term movements.

Subsequent to the above operations and prior to the placement of footings, structural site grading fill, floor slabs, or pavements, the exposed natural subgrade must be proofrolled by passing moderate-weight rubber tire-mounted construction equipment over the surface at least twice. If any loose, soft, or disturbed zones are encountered, they must be completely removed in footing and floor slab areas and replaced with granular structural fill. If removal depth required is greater than two feet, G^2 must be notified to provide further recommendations. In pavement areas, unsuitable soils encountered during recompaction and proofrolling must be removed to a maximum depth of two feet and replaced with compacted granular structural fill.

Surface vegetation and other deleterious materials should generally be removed from the site. Topsoil if encountered, although unsuitable for re-utilization as structural fill, may be stockpiled for subsequent landscaping purposes.

5.2.2 Temporary Excavations

Temporary construction excavations through fine-grained cohesive soils, not exceeding four feet in depth, above or below the groundwater table, may be constructed with near-vertical sideslopes. Temporary excavations up to eight feet deep in fine-grained cohesive soils and above or below the water table may be constructed with sideslopes no steeper than one-half horizontal to one vertical (0.5H:1.0V). Temporary excavations up to eight feet deep in granular soils above or below the water table may be constructed with sideslopes no steeper than one horizontal to one vertical (1.0H:1.0V).

Utility trench excavations must conform within Occupational Safety and Health (OSHA) guidelines for trench safety.

To minimize disturbance to the underlying soils, it is our recommendation that footings be excavated with a backhoe equipped with a smooth-lip bucket.

All excavations must be inspected periodically by qualified personnel. If any signs of instability or excessive sloughing are noted, immediate remedial action must be initiated.

5.2.3 Structural Fill

Structural fill is defined as all fill which will ultimately be subjected to structural loadings, such as imposed by footings, floor slabs, pavements, etc. Structural fill will be required as backfill over foundations and utilities, as site grading fill, and in some areas, as replacement fill below footings. It is recommended that all structural fill must be free of sod, rubbish, topsoil, frozen soil, and other deleterious materials. Structural site grading fill is defined as fill placed over fairly large open areas to raise the overall site grade. For structural site grading fill, the maximum particle size should generally not exceed four inches; although, occasional larger particles, not exceeding six inches in diameter may be incorporated if placed randomly in a manner such that “honeycombing” does not occur and the desired degree of compaction can be achieved. The maximum particle size within structural fill placed within confined areas should generally be restricted to two inches.

The on-site natural soils may be utilized as structural site grading fill. It should be noted that unless moisture control is maintained, utilization of fine-grained soils (if encountered) as structural site grading fill <u>will be very difficult, if not impossible</u>, during wet and cold periods of the year. Only granular soils are recommended as structural fill in confined areas, such as around foundations and within utility trenches.

To stabilize soft subgrade conditions a mixture of coarse gravels and cobbles and/or one and one-half- to two-inch gravel (stabilizing fill) should be utilized.

Non-structural site grading fill is defined as all fill material not designated as structural fill and may consist of any cohesive or granular soils not containing excessive amounts of degradable material.

5.2.4 Fill Placement and Compaction

Coarse gravel and cobble mixtures (stabilizing fill), if utilized, shall be end-dumped, spread to a maximum loose lift thickness of 15 inches, and compacted by dropping a backhoe bucket onto the surface continuously at least twice. As an alternative, the fill may be compacted by passing moderately heavy construction equipment or large self-propelled compaction equipment at least twice. Subsequent fill material placed over the coarse gravels and cobbles shall be adequately placed so that the “fines” are “worked into” the voids in the underlying coarser gravels and cobbles.

Structural fill shall be placed in lifts not exceeding eight inches in loose thickness. Structural fills shall be compacted in accordance with the percent of the maximum dry density as determined by the AASHTO[1] T-180 (ASTM[2] D-1557) compaction criteria in accordance with the following table:

Location	Total Fill Thickness (feet)	Minimum Percentage of Maximum Dry Density
Beneath an area extending at least 3 feet beyond the perimeter of the structure	0 to 8	95
Outside area defined above	0 to 5	90
Outside area defined above	5 to 8	92
Road base	-	96

Structural fills greater than eight feet thick are not anticipated at the site.

Subsequent to stripping and prior to the placement of structural site grading fill, the subgrade must be prepared as discussed in Section 5.2.1, Site Preparation, of this report. In confined areas, subgrade preparation should consist of the removal of all loose or disturbed soils.

1 American Association of State Highway and Transportation Officials
2 American Society for Testing and Materials

Non-structural fill may be placed in lifts not exceeding 12 inches in loose thickness and compacted by passing construction, spreading, or hauling equipment over the surface at least twice.

5.2.5 Utility Trenches

All utility trench backfill material below structurally loaded facilities (flatwork, floor slabs, roads, etc.) should be placed at the same density requirements established for structural fill. If the surface of the backfill becomes disturbed during the course of construction, the backfill should be proofrolled and/or properly compacted prior to the construction of any exterior flatwork over a backfilled trench. Proofrolling may be performed by passing moderately loaded rubber tire-mounted construction equipment uniformly over the surface at least twice. If excessively loose or soft areas are encountered during proofrolling, they should be removed to a maximum depth of two feet below design finish grade and replaced with structural fill.

Most utility companies and City-County governments are now requiring that Type A-1 or A-1-a (AASHTO Designation – basically granular soils with limited fines) soils be used as backfill over utilities. These organizations are also requiring that in public roadways the backfill over major utilities be compacted over the full depth of fill to at least 96 percent of the maximum dry density as determined by the AASHTO T-180 (ASTM D-1557) method of compaction. We recommend that as the major utilities continue onto the site that these compaction specifications are followed.

The natural fine-grained cohesive soils (if encountered) are not recommended for use as trench backfill.

5.2.6 Areal Settlements

Areal settlements resulting from site grading fills as much as three feet should be less than one inch. These settlements are in addition to settlements induced by foundation and floor slab loads. To reduce the total settlement that the structure will realize, site grading fill must be placed as far in advance of other construction as possible. The majority of this settlement will occur during placement.

5.3 SPREAD AND CONTINUOUS WALL FOUNDATIONS

5.3.1 Design Data

The proposed structure may be supported upon conventional spread and continuous wall foundations. The footings may be established on suitable, undisturbed natural soils and/or structural fills extending to suitable undisturbed natural soils.

For design, the following parameters are provided:

Minimum Recommended Depth of Embedment for Frost Protection	- 30 inches
Minimum Recommended Depth of Embedment for Non-frost Conditions	- 15 inches
Recommended Minimum Width for Continuous Wall Footings	- 18 inches
Minimum Recommended Width for Isolated Spread Footings	- 24 inches
Recommended Net Bearing Pressures for Real Load Conditions	
For footings on suitable natural soils and/or structural fill extending to suitable natural soils	2,500 pounds per square foot
Bearing Pressure Increase for Seismic Loading	- 50 percent*

* Not applicable for edge bearing pressure when the footings are established upon granular soil.

The term “net bearing pressure” refers to the pressure imposed by the portion of the structure located above lowest adjacent final grade. Therefore, the weight of the footing and backfill to lowest adjacent final grade need not be considered. Real loads are defined as the total of all dead plus frequently applied live loads. Total load includes all dead and live loads, including seismic and wind.

5.3.2 Installation

Under no circumstances should the footings be installed overlying non-engineered fills, upon soft or disturbed soils, loose surficial soils, construction debris, frozen soil, or within ponded water.

If the natural soils upon which the footings are to be established become loose or disturbed, they must be removed and replaced with granular structural fill. If granular structural fill upon which the footings are to be established becomes disturbed, they should be recompacted to the requirements for structural fill or be removed and replaced with structural fill.

The width of structural replacement fill, as required below more heavily loaded footings, should be extended laterally at least one foot beyond the edges of the footings in all directions for each foot of fill thickness beneath the footings.

5.3.3 Settlements

Settlements of foundations designed and installed in accordance with above recommendations and supporting maximum projected structural loads are anticipated to be less than one inch. Settlements are expected to occur rapidly with approximately 60 to 70 percent of the settlements occurring during construction.

5.4 LATERAL RESISTANCE

Lateral loads imposed upon foundations due to wind or seismic forces may be resisted by the development of passive earth pressures and friction between the base of the footings and the supporting soils. In determining frictional resistance, a coefficient of 0.40 should be utilized for the natural fine-grained soils. In determining frictional resistance, a coefficient of 0.45 should be utilized for the natural granular soils. Passive resistance provided by properly placed and compacted granular structural fill above the water table may be considered equivalent to a fluid with a density of 300 pounds per cubic foot. Below the water table, this granular soil should be considered equivalent to a fluid with a density of 150 pounds per cubic foot.

A combination of passive earth resistance and friction may be utilized provided that the friction component of the total is divided by 1.5.

5.5 LATERAL PRESSURES

The lateral pressure parameters as presented within this section, assume that the backfill will consist of a drained granular soil placed and compacted in accordance with the recommendations presented herein. The lateral pressures imposed upon subgrade facilities will, therefore, be basically dependent upon the relative rigidity and movement of the backfilled structure. For active walls, such as retaining walls which can move outward (away from the backfill), granular backfill may be considered equivalent to a fluid with a density of 35 pounds per cubic foot in computing lateral pressures. For more rigid basement walls that are not more than 10 inches thick and 12 feet or less in height, granular backfill may be considered equivalent to a fluid with a density of 45 pounds per cubic foot. For very rigid non-yielding walls, granular backfill should be considered equivalent to a fluid with a density of at least 60 pounds per cubic foot. The above values assume that the surface of the soils slope behind the wall is horizontal, that the granular fill has been placed and lightly compacted, not as a structural fill. If the fill is placed as a structural fill, the values should be increased to 45 pounds per cubic foot, 60 pounds per cubic foot, and 120 pounds per cubic foot, respectively. If the slope behind the wall is two horizontal to one vertical the values for purely active walls and basement walls should increase to 57 pounds per cubic foot and 67 pounds per cubic foot, respectively.

Recommended average lateral uniform pressure for various height walls are tabulated below and assume a granular wall backfill with a horizontal grade above the wall. It should be noted that the lateral pressures as quoted assume that the backfill materials will not become saturated. If the backfill becomes saturated, the above values may be decreased by one-half; however, full hydrostatic water pressures will have to be included.

Wall Height (feet)	Uniform Seismic Lateral Pressure* (psf)
4	56

* Maximum short-term pressures, they are not sustained loads.

Note that the pressures presented in this section do not include surcharge loadings, such as floor slabs, adjacent footings, etc.

5.6 FLOOR SLABS

Floor slabs may be established upon structural fill extending to suitable undisturbed natural soils. Loose/disturbed surficial soils, non-engineered fills and topsoil are not considered suitable. To provide a capillary break, it is recommended that floor slabs be directly underlain by at least four inches of “free-draining” fill, such as “pea” gravel or three-quarters- to one-inch minus clean gap-graded gravel. Settlements of lightly to moderately loaded floor slabs are anticipated to be minor.

5.7 PAVEMENTS

The properly prepared non-engineered fills will exhibit poor engineering characteristics when saturated or nearly saturated. Loose/disturbed surficial soil/non-engineered fills may remain in flexible pavement areas if properly prepared, as stated previously in this report. Rigid pavements shall not be placed overlying loose/disturbed surficial soil/non-engineered fills, even if properly prepared. Considering the loose/disturbed surficial soil/existing non-engineered soils as the subgrade soils and the projected traffic, the pavement sections on the following pages are recommended.

<u>Parking Areas</u>

(Light Volume of Automobiles and Light Trucks,
Occasional Medium-Weight Trucks,
and No Heavy-Weight Trucks)
[1 equivalent 18-kip axle load per day]

<u>Flexible:</u>

2.5 inches	Asphalt concrete
8.0 inches	Aggregate base
Over	Properly prepared natural soils, properly prepared existing non-engineered fill, and/or structural site grading fill extending to suitable stabilized natural soils.

<u>Rigid:</u>

5.0 inches	Portland cement concrete (non-reinforced)
4.0 inches	Aggregate base
Over	Properly prepared natural soils, and/or structural site grading fill extending to suitable stabilized natural soils.*

* Rigid pavements shall not be placed over non-engineered fills, even if properly prepared.

Primary Roadway Areas

(Moderate Volume of Automobiles and Light Trucks,
Light Volume of Medium-Weight Trucks,
and Occasional Heavy-Weight Trucks)
[5 equivalent 18-kip axle loads per day]

Flexible:

3.0 inches	Asphalt concrete
8.0 inches	Aggregate base
Over	Properly prepared natural soils, properly prepared existing non-engineered fill, and/or structural site grading fill extending to suitable stabilized natural soils.

Rigid:

5.5 inches	Portland cement concrete (non-reinforced)
5.0 inches	Aggregate base
Over	Properly prepared natural soils, and/or structural site grading fill extending to suitable stabilized natural soils.*

* Rigid pavements shall not be placed over non-engineered fills, even if properly prepared.

For dumpster pads, we recommend a pavement section consisting of six and one-half inches of Portland cement concrete, four inches of aggregate base, over properly prepared natural stabilized subgrade or site grading structural fills.

These above rigid pavement sections are for non-reinforced Portland cement concrete. Concrete should be designed in accordance with the American Concrete Institute (ACI) and joint details should conform to the Portland Cement Association (PCA) guidelines. The concrete should have a minimum 28-day unconfined compressive strength of 4,000 pounds per square inch and contain 6 percent ±1 percent air-entrainment.

5.8 GEOSEISMIC SETTING

5.8.1 General

In July 2023, the State of Utah adopted the 2021 International Building Code (IBC). The IBC 2021 code determines the seismic hazard for a site based upon 2014 mapping of bedrock accelerations prepared by the United States Geologic Survey (USGS) and the soil site class. The USGS values are presented on maps incorporated into the IBC code and are also available based on latitude and longitude coordinates (grid points).

The structure must be designed in accordance with the procedure presented in Section 1613, Earthquake Loads, of the IBC 2021 edition.

5.8.2 Faulting

Based on our review of available literature, no active faults pass through or immediately adjacent to the site.

5.8.3 Soil Class

Based on the soils encountered and our knowledge of the underlying geology, we recommend that Site Class D - Stiff Soil Profile as defined in Table 20.3-1, Site Classification, of ASCE 7-16 be utilized for dynamic structural analysis.

5.8.4 Ground Motions

The IBC 2021 code is based on 2014 USGS mapping, which provides values of short and long period accelerations for the Site Class B boundary for the Maximum Considered Earthquake (MCE). This Site Class B boundary represents a hypothetical sandstone bedrock surface and must be corrected for local soil conditions. The following table summarizes the peak ground and short and long period accelerations for a MCE event and incorporates a soil amplification factor for a Site Class D soil profile in the second column. Based on the site latitude and longitude (41.6554 degrees north and -111.8180 degrees west, respectively), the values for this site are tabulated on the following page.

Spectral Acceleration Value, T Seconds	Site Class B-C Boundary [mapped values] (% g)	Site Class D [adjusted for site class effects] (% g)
Peak Ground Acceleration (Geo-Mean)	44.1	51.1
0.2 Seconds (Short Period Acceleration)	S_S = 101.7	S_{MS} = 111.2
1.0 Seconds (Long Period Acceleration)	S_1 = 34.2	S_{M1} = *

* See Section 11.4.8 for requirements on site-specific ground motion studies. Please contact us for a proposal, if needed.

5.8.5 Liquefaction

The site is located in an area that has been identified by the Utah Geological Survey as having “very low” liquefaction potential. Liquefaction is defined as the condition when saturated, loose, finer-grained sand-type soils lose their support capabilities because of excessive pore water pressure which develops during a seismic event.

Due to the absence of groundwater to the depths explored our analysis indicates liquefaction is not anticipated during the design seismic event.

5.9 SITE OBSERVATIONS

As previously mentioned, loose/disturbed surficial soil and potential for encountering non-engineered fills are present across the surface of the site to varying depths. Therefore, we recommend that a qualified geotechnical engineer from our staff observe the foundation excavations to identify that these soils have been removed and that suitable soils have been encountered below structure.

We appreciate the opportunity of providing this service for you. If you have any questions or require additional information, please do not hesitate to contact us.

Respectfully submitted,

Gordon Geotechnical Engineering, Inc.

Joshua M. Whitney, State of Utah No. 6252902
Senior Engineer

Reviewed by:

Patrick R. Emery, State of Utah No. 7941710
Senior Engineer

JMW/PRE:sn

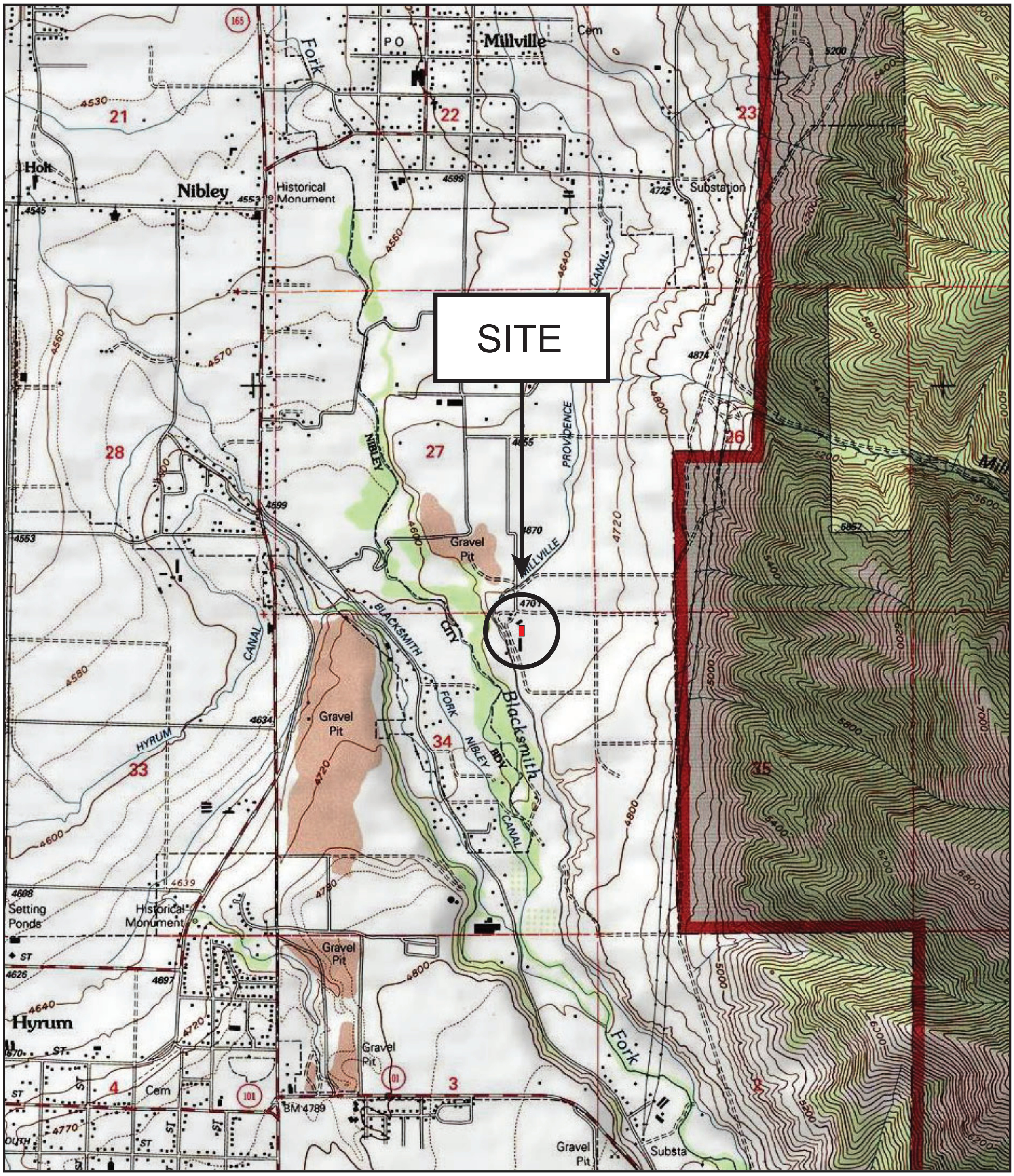
Encl. Figure 1, Vicinity Map
Figure 2, Area Map
Figures 3A and 3B, Log of Test Pits
Figure 4, Unified Soil Classification System
Figure 5, Photographs

Addressee (3 + email)

PROFESSIONAL ENGINEER
7941710
PATRICK R.
EMERY
11-13-24
STATE OF UTAH

CUSHING TERRELL
JOB NO. 1342-001-24

GORDON GEOTECHNICAL ENGINEERING, INC.



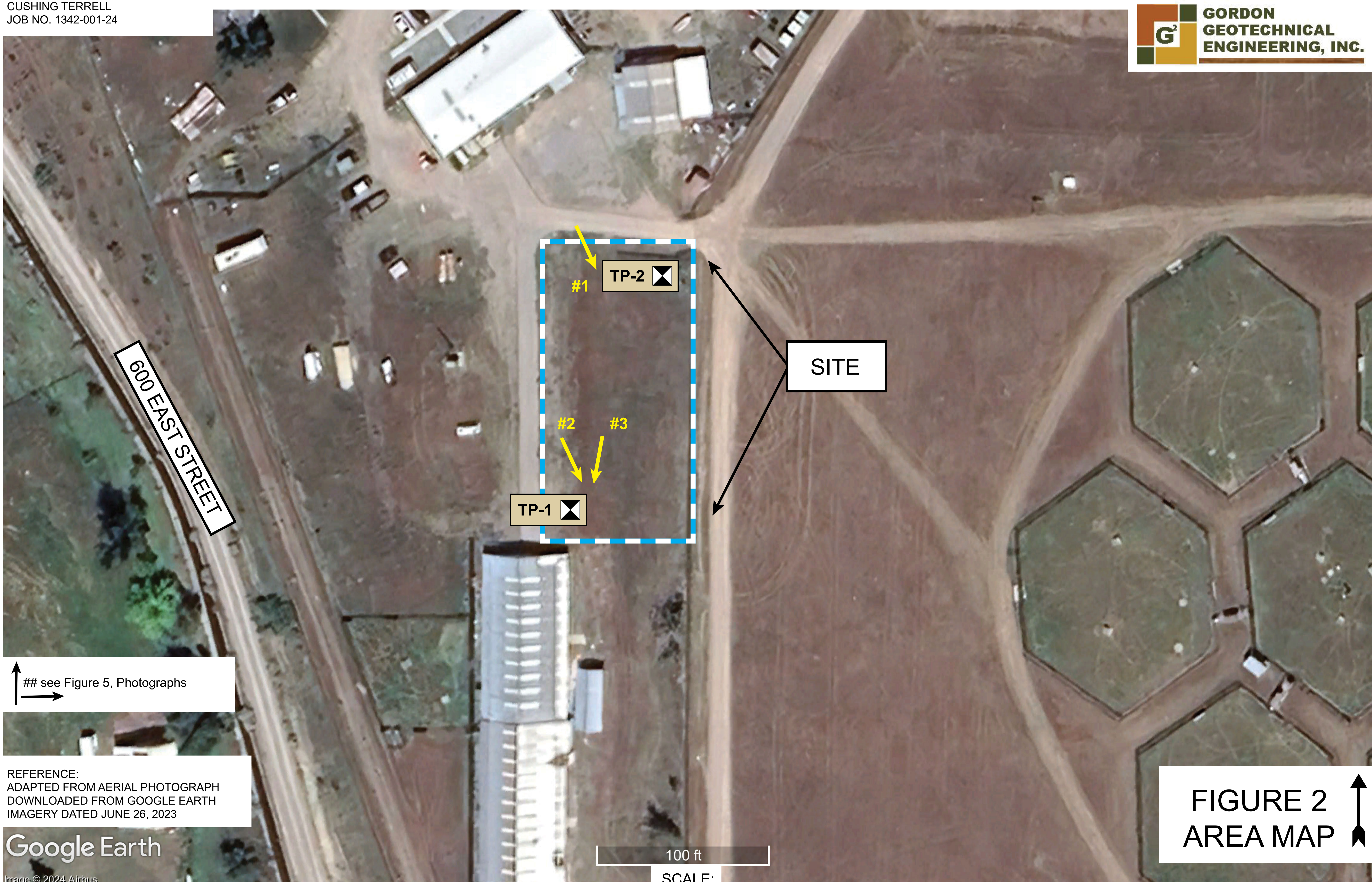
SCALE IN FEET
1000 0 1000 2000

FIGURE 1
VICINITY MAP

REFERENCE:
USGS 7.5 MINUTE TOPOGRAPHIC QUADRANGLE MAP
TITLED "LOGAN, UTAH", DATED 1998

CUSHING TERRELL
JOB NO. 1342-001-24

GORDON GEOTECHNICAL ENGINEERING, INC.



REFERENCE:
ADAPTED FROM AERIAL PHOTOGRAPH
DOWNLOADED FROM GOOGLE EARTH
IMAGERY DATED JUNE 26, 2023

FIGURE 2
AREA MAP

Gordon Geotechnical Engineering, Inc.
4426 South Century Drive, Suite 100
Salt Lake City, Utah 84123

TEST PIT TP-1

Project Name: Proposed Millville Kennel
Project No.: 1342-001-24
Location: 4200 South 600 East, Millville, Utah
Client: Cushing Terrell
Excavating Method: Case 580 Backhoe
Date Excavated: 11-01-24
Elevation: ---
Water Level: No groundwater encountered.
Remarks:

DESCRIPTION	GRAPHIC LOG	WATER LEVEL	DEPTH (FT.)	SAMPLE SYMBOL	SAMPLE TYPE	BLOWS/FT.	MOISTURE (%)	DRY DENSITY (PCF)	%PASSING 200	LIQUID LIMIT (%)	PLASTIC LIMIT (%)	REMARKS
Ground Surface			0									
FINE AND COARSE GRAVEL with fine to coarse sand, cobbles/boulders, and some silt; major roots (topsoil) to 4"; brown (GP/GM)												slightly moist "medium dense"
					B		1.8		6.2			
			5									
FINE AND COARSE GRAVEL with some fine to coarse sand, cobbles, boulders, and trace silt; brown (GP)												
					B		1.1		2.2			
			10		B							
			15									
			20									
			25									

Stopped excavating at 10.5'.

No groundwater encountered at time of excavating.

No significant sidewall caving.

The discussion in the text under the section titled, SUBSURFACE CONDITIONS, is necessary for proper understanding of the nature of the subsurface material.

FIGURE 3A

Gordon Geotechnical Engineering, Inc.
4426 South Century Drive, Suite 100
Salt Lake City, Utah 84123

TEST PIT TP-2

Project Name: Proposed Millville Kennel

Project No.: 1342-001-24

Location: 4200 South 600 East, Millville, Utah

Client: Cushing Terrell

Excavating Method: Case 580 Backhoe

Date Excavated: 11-01-24

Elevation: ---

Water Level: No groundwater encountered.

Remarks:

DESCRIPTION	GRAPHIC LOG	WATER LEVEL	DEPTH (FT.)	SAMPLE SYMBOL	SAMPLE TYPE	BLOWS/FT.	MOISTURE (%)	DRY DENSITY (PCF)	%PASSING 200	LIQUID LIMIT (%)	PLASTIC LIMIT (%)	REMARKS
Ground Surface												
FINE AND COARSE GRAVEL/FINE TO COARSE SAND with cobbles/boulders and trace silt; major roots (topsoil) to 5"; brown (GP/SP)			0									slightly moist "medium dense"
					B		1.2		4.7			
			5									
					B							
Stopped excavating at 7.5'.												
No groundwater encountered at time of excavating.			10									
No significant sidewall caving.												
			15									
			20									
			25									

The discussion in the text under the section titled, SUBSURFACE CONDITIONS, is necessary for proper understanding of the nature of the subsurface material.

FIGURE 3B

GORDON GEOTECHNICAL ENGINEERING, INC.

UNIFIED SOIL CLASSIFICATION SYSTEM

FIELD IDENTIFICATION PROCEDURES							GRAPH SYMBOL	LETTER SYMBOL [1]	TYPICAL DESCRIPTIONS
COARSE GRAINED SOILS More than half of material is larger than No. 200 sieve size. [2] (The No. 200 sieve size is about the smallest particle visible to the naked eye)	GRAVELS More than half of coarse fraction is larger than No. 4 sieve size. (For visual classifications, the 1/4" size may be used as equivalent to the No. 4 sieve size.)	CLEAN GRAVELS (Little or no fines)	Wide range in grain size and substantial amounts of all intermediate particle sizes.					GW	Well graded gravels, gravel-sand mixtures, little or no fines.
			Predominantly one size or a range of sizes with some intermediate sizes missing.					GP	Poorly graded gravels, gravel-sand mixtures, little or no fines.
		GRAVELS WITH FINES (Appreciable amount of fines)	Non-plastic fines (for identification procedures see ML below).					GM	Silty gravels, poorly graded gravel-sand-silt mixtures.
			Plastic fines (for identification procedures see CL below).					GC	Clayey gravels, poorly graded gravel-sand-clay mixtures.
	SANDS More than half of coarse fraction is smaller than No. 4 sieve size. (For visual classifications, the 1/4" size may be used as equivalent to the No. 4 sieve size.)	CLEAN SANDS (Little or no fines)	Wide range in grain sizes and substantial amounts of all intermediate particle sizes.					SW	Well graded sands, gravelly sands, little or no fines.
			Predominantly one size or a range of sizes with some intermediate sizes missing.					SP	Poorly graded sands, gravelly sands, little or no fines.
		SANDS WITH FINES (Appreciable amount of fines)	Non-plastic fines (for identification procedures see ML below).					SM	Silty sands, poorly graded sand-silt mixtures.
			Plastic fines (for identification procedures see CL below).					SC	Clayey sands, poorly graded sand-clay mixtures.
	IDENTIFICATION PROCEDURES ON FRACTION SMALLER THAN No. 40 SIEVE SIZE								
			DRY STRENGTH (CRUSHING CHARACTERISTICS)	DILATANCY (REACTION TO SHAKING)	TOUGHNESS (CONSISTENCY NEAR PLASTIC LIMIT)				
FINE GRAINED SOILS More than half of material is smaller than No. 200 sieve size. (The No. 200 sieve size is about the smallest particle visible to the naked eye)	SILTS AND CLAYS Liquid limit less than 50		None to slight	Quick to slow	None			ML	Inorganic silts and very fine sands, rock flour, silty or clayey fine sand with slight plasticity.
			Medium to high	None to very slow	Medium			CL	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays.
			Slight to medium	Slow	Slight			OL	Organic silts and organic silt-clays of low plasticity.
	SILTS AND CLAYS Liquid limit greater than 50		Slight to medium	Slow to none	Slight to medium			MH	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts.
			High to very high	None	High			CH	Inorganic clays of high plasticity, fat clays.
			Medium to high	None to very slow	Slight to medium			OH	Organic clays of medium to high plasticity.
HIGHLY ORGANIC SOILS			Readily identified by color, odor, spongy feel and frequently by fibrous texture.					Pt	Peat and other highly organic soils.

[1] Boundary classifications: —Soils possessing characteristics of two groups are designated by combinations of group symbols. For example GW—GC, well graded gravel—sand mixture with clay binder.
[2] All sieve sizes on this chart are U.S. standard.

GENERAL NOTES

1. In general, Unified Soil Classification Designations presented on the logs were evaluated by visual methods only. There rore, actual designations (based on laboratory testing) may differ.

2. Lines seperating strata on the logs represent approximate boundaries only Actual transitions may be gradual.

3. Logs represent general soil conditions observed at teh point of exploration onthe date indicated.

4. No warranty is provided as to the continuity of soil conditions between individual sample locations.

FINE - GRAINED SOIL

CONSISTENCY	SPT (blows/ft)	TORVANE UNDRAINED SHEAR STRENGTH (tsf)	POCKET PENETROMETER UNCONFINED COMPRESSIVE STRENGTH (tsf)	FIELD TEST
Very Soft	<2	<0.125	<0.25	Easily penetrated several inches by Thumb. Squeezes through fingers.
Soft	2 - 4	0.125 - 0.25	0.25 - 0.5	Easily penetrated 1 " by Thumb . Molded by light finger pressure.
Medium Stiff	4 - 8	0.25 - 0.5	0.5 - 1.0	Penetrated over 1/2 " by Thumb with moerate effort. Molded by strong finger pressure.
Stiff	8 - 15	0.5 - 1.0	1.0 - 2.0	Indented about 1/2 " by Thumb but penetrated only with great effort
Very Stiff	15 - 30	1.0 - 2.0	2.0 - 4.0	Readily indented by Thumbnail
Hard	>30	>2.0	>4.0	Indented with difficulty by Thumbnail

LOG KEY SYMBOLS

Bulk / Bag Sample	Thin Wall
Standard Penetration Split Spoon Sampler	No Recovery
Rock Core	3-3/4" ID D&M Sampler
	3" ID D&M Sampler
Water Level	California Sampler

COARSE -GRAINDE SOIL

APPERENT DENSITY	SPT (blows/ft)	RELATIVE DENSITY (%)	FIELD TEST
Very Loose	<4	0 - 15	Easily penetrated with 1/2 " reinforcing rod pushed by hand
Loose	4 - 10	15 - 35	Difficult to penetrated with 1/2 " reinforcing rod pushed by hand
Medium Dense	10 - 30	35 - 65	Easily penetrated a foot with 1/2 " reinforcing rod driven with 5-lb hammer
Dense	30 - 50	65 - 85	Difficult to penetrated a foot with 1/2 " reinforcing rod driven with 5-lb hammer
Very Dense	>50	85 - 100	Penetrated only a few inches with 1/2 " reinforcing rod driven with 5-lb hammer

STRATIFICATION

DESCRIPTION	THICKNESS
SEAM	1/16 - 1/2 "
LAYER	1/2 - 12 "

DESCRIPTION	THICKNESS
Occasional	One or less per foot of thickness
Frequent	More than on per foot of thichness

CEMENTATION

DESCRIPTION	DESCRIPTION
Weakely	Crumbles or breaks with handling of slight finger pressure
Moderately	Crumbles or breaks with considerable finger pressure
Strongly	Will not crumbles or breaks with finger pressure

MODIFIERS

DESCRIPTION	%
Trace	<5
Some	5 - 12
With	>12

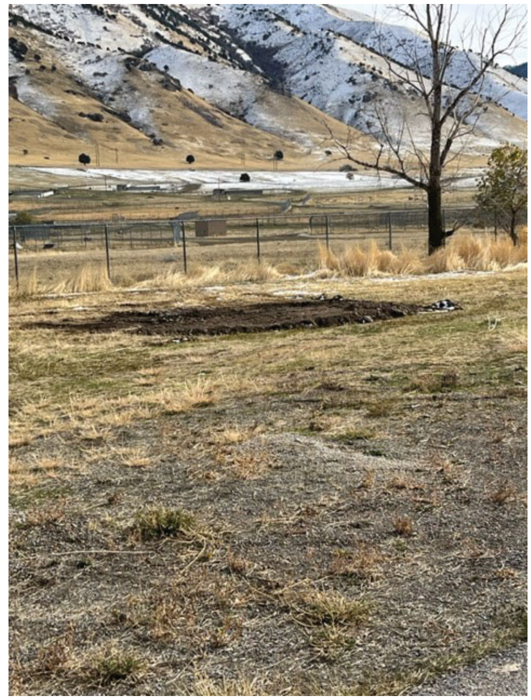
MOISTURE CONTENT

DESCRIPTION	FIELD TEST
Dry	Absence of moisture, dusty, dry to the touch
Moist	Damp but no visible water
Wet	Visible water, usually soil below Water Table

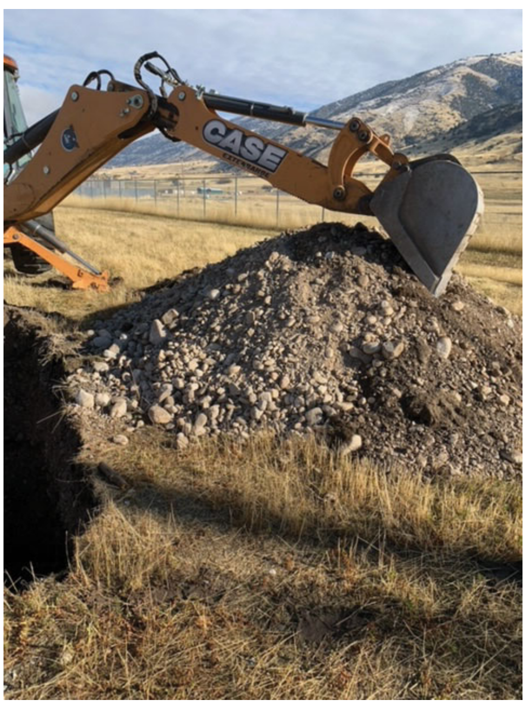
FIGURE 4

GORDON GEOTECHNICAL ENGINEERING, INC.

#1 Looking southeast.



#2 Looking southeast.



#3 Looking at Test Pit TP-1.

FIGURE 5
PHOTOGRAPHS

Locations and direction, see Figure 2, Area Map